

MolPallette — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallette/coconut-flavordb_full_r050/naveja_recap

generated 2026-08-20 06:32

A • Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.5, 'include_ring': True}
core_ratio	0.5
max_cores	10
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	2
rdkit_version	2026.03.2
created_at	2026-08-20T05:47:51

B · Composition

quantity	value
molecules	413,459
from coconut	391,965 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	3,389,194
per molecule	mean 8.20, median 10, max 10
R-group slots	3,670,468
per decomposition	mean 1.08
k >= 2	6.69% of decompositions

k>=2 is what the islinked subset space and the core-decoration objective have to work with.

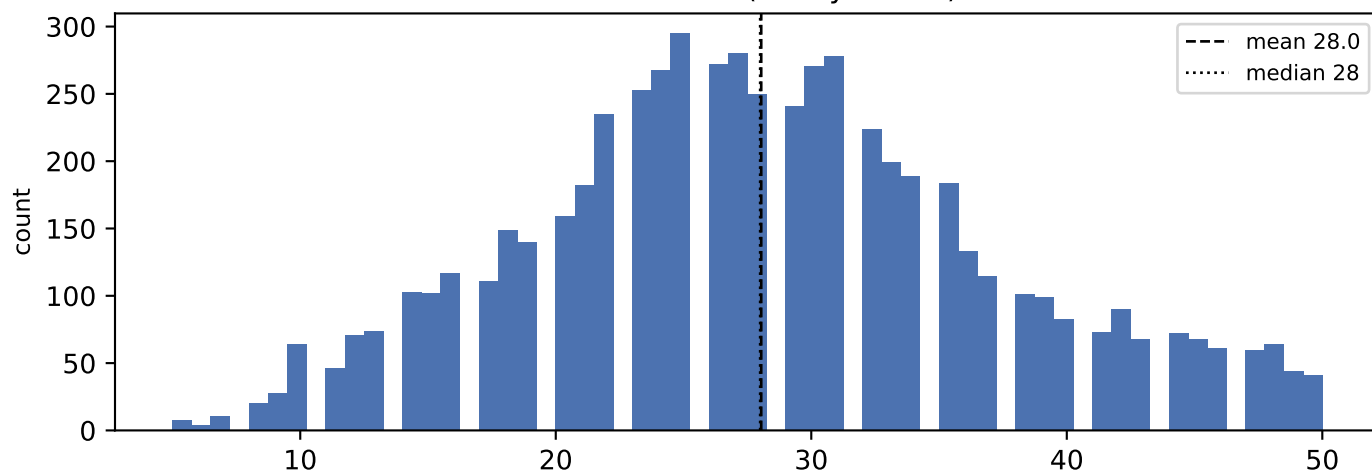
C · Augmentation space

scheme	size
sampling_unit = molecule	413,459 instances/epoch
sampling_unit = decomposition	3,389,194 instances/epoch (8.2x)
full (mol, core, islinked) space	4,150,830 (10.0 per molecule)

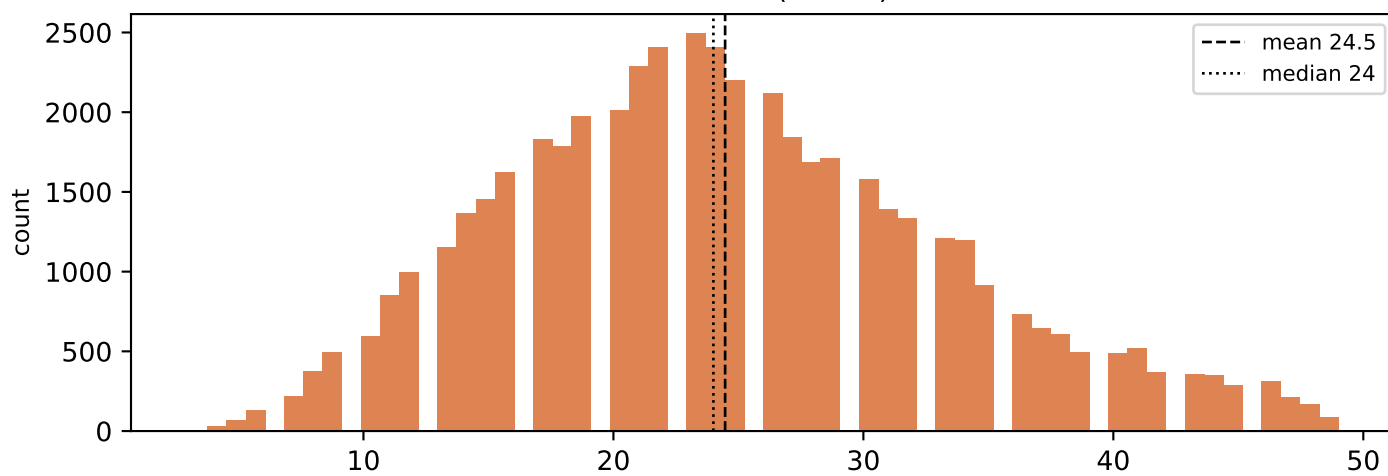
MolPLA's two stages: one molecule yields many cores; one core yields $2^k - 1$ islinked subsets.

D • Size distributions

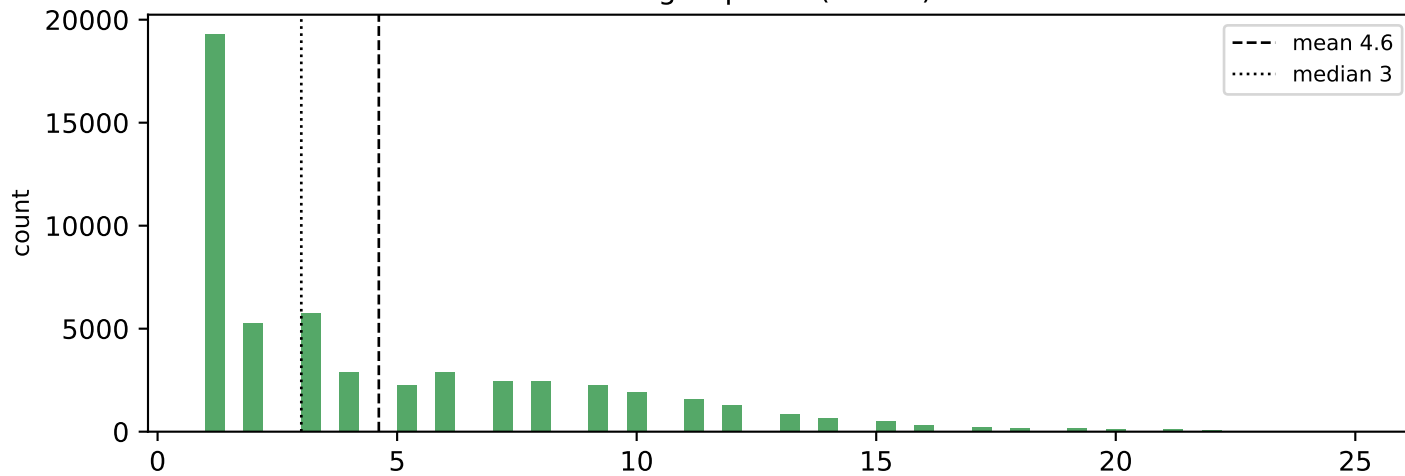
molecule size (heavy atoms)



core size (atoms)

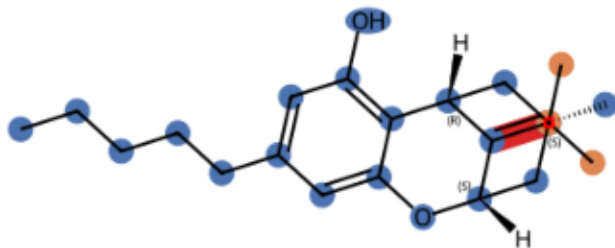


R-group size (atoms)

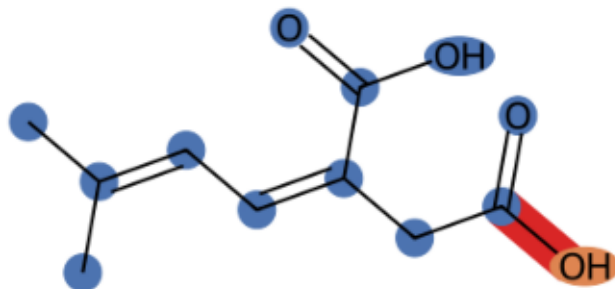


E · Example decompositions (ratio 0.5)

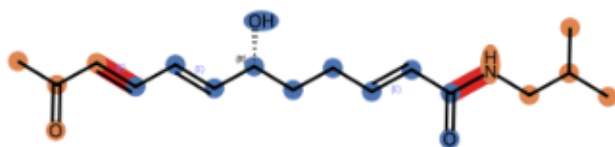
CNP0000074 k=1 core 20 atoms, R-groups [3]



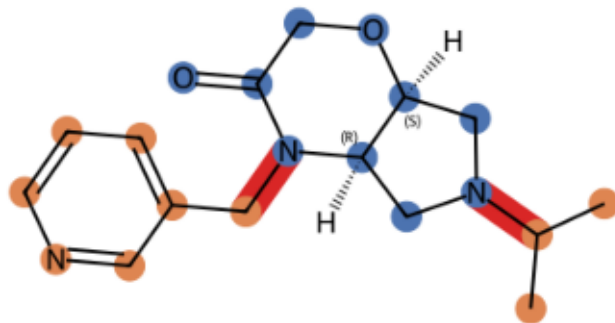
CNP0000161 k=1 core 12 atoms, R-groups [1]



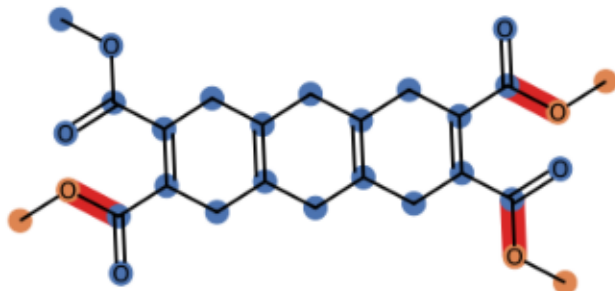
CNP0000307 k=2 core 11 atoms, R-groups [4, 5]



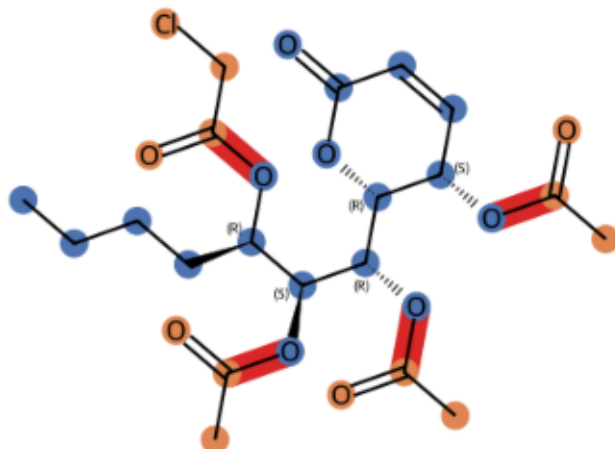
CNP0003697 k=2 core 10 atoms, R-groups [3, 7]



CNP0025561 k=3 core 24 atoms, R-groups [2, 2, 2]

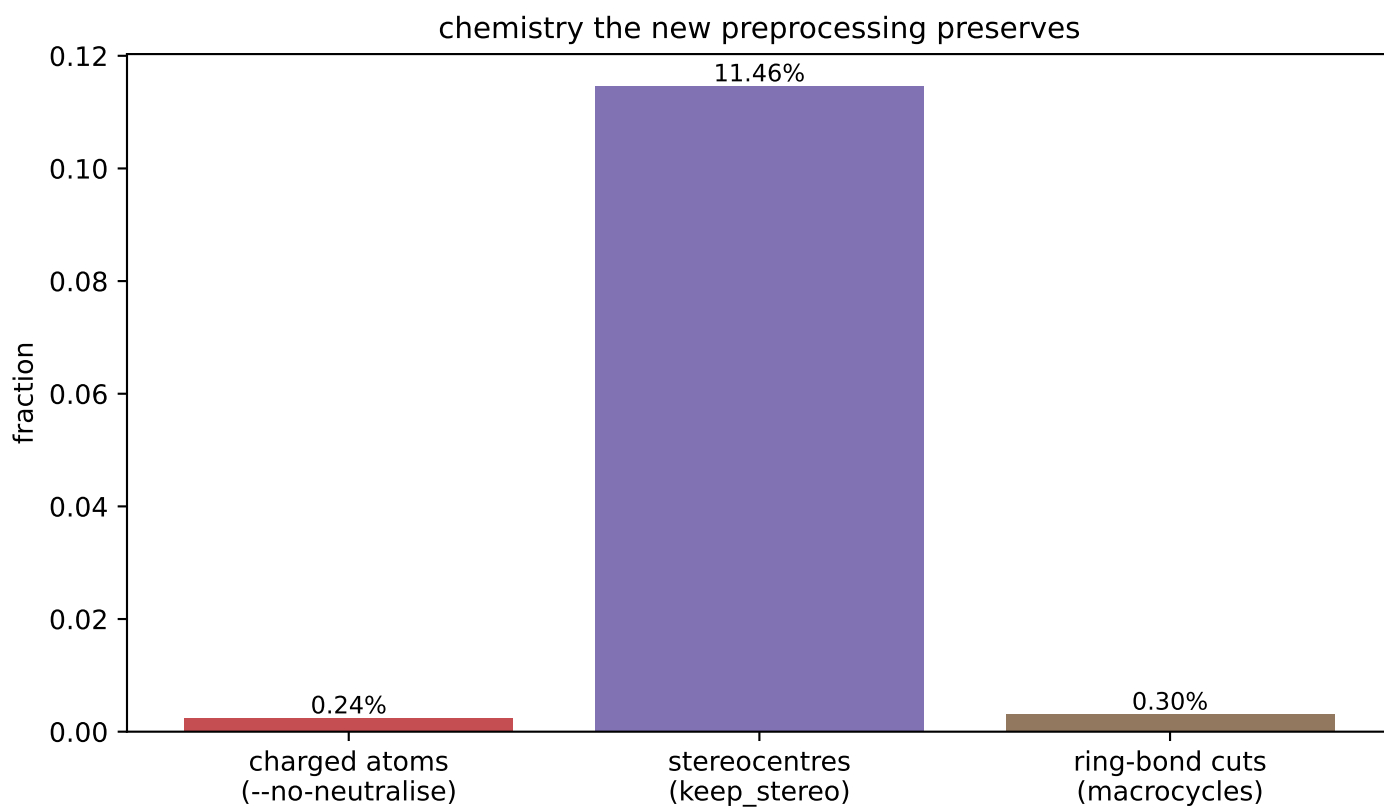
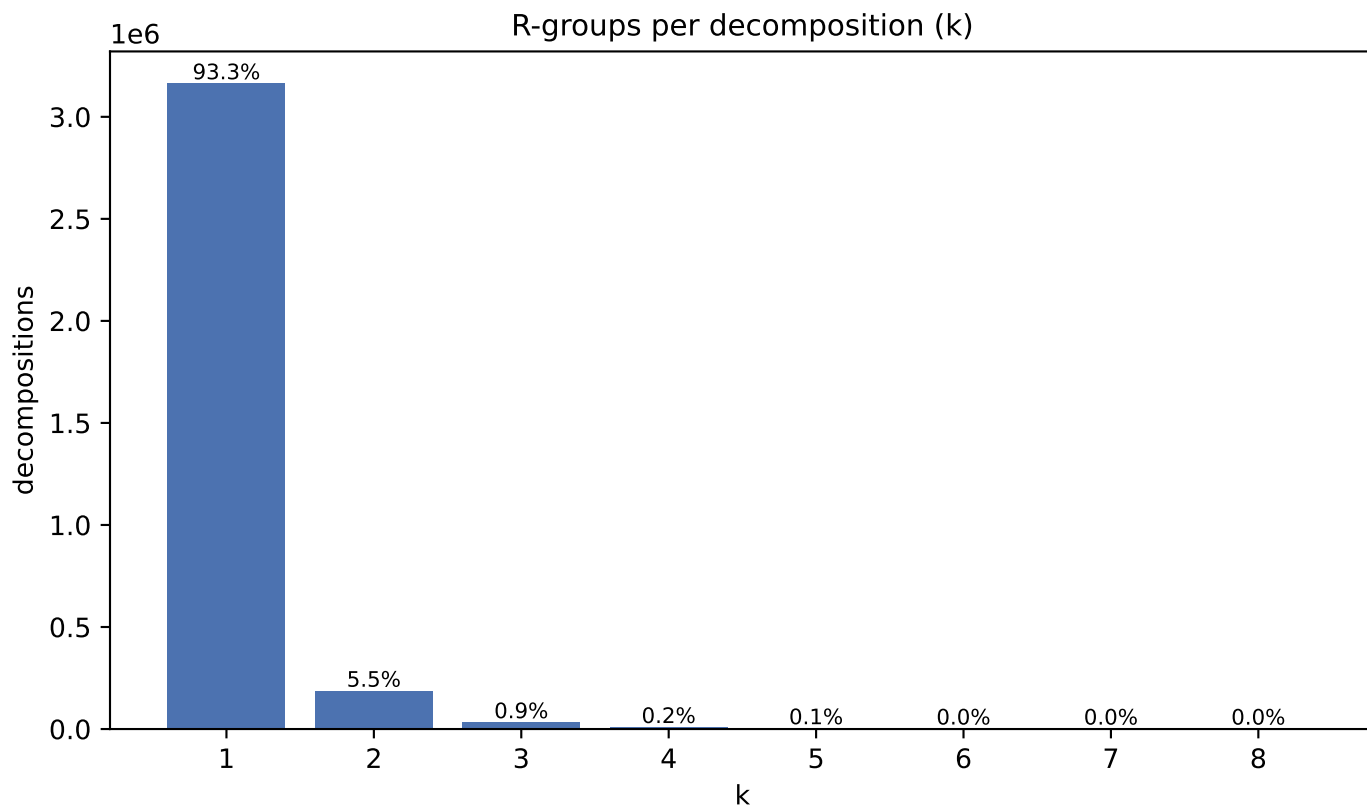


CNP0078333 k=4 core 18 atoms, R-groups [4, 3, 3, 3]

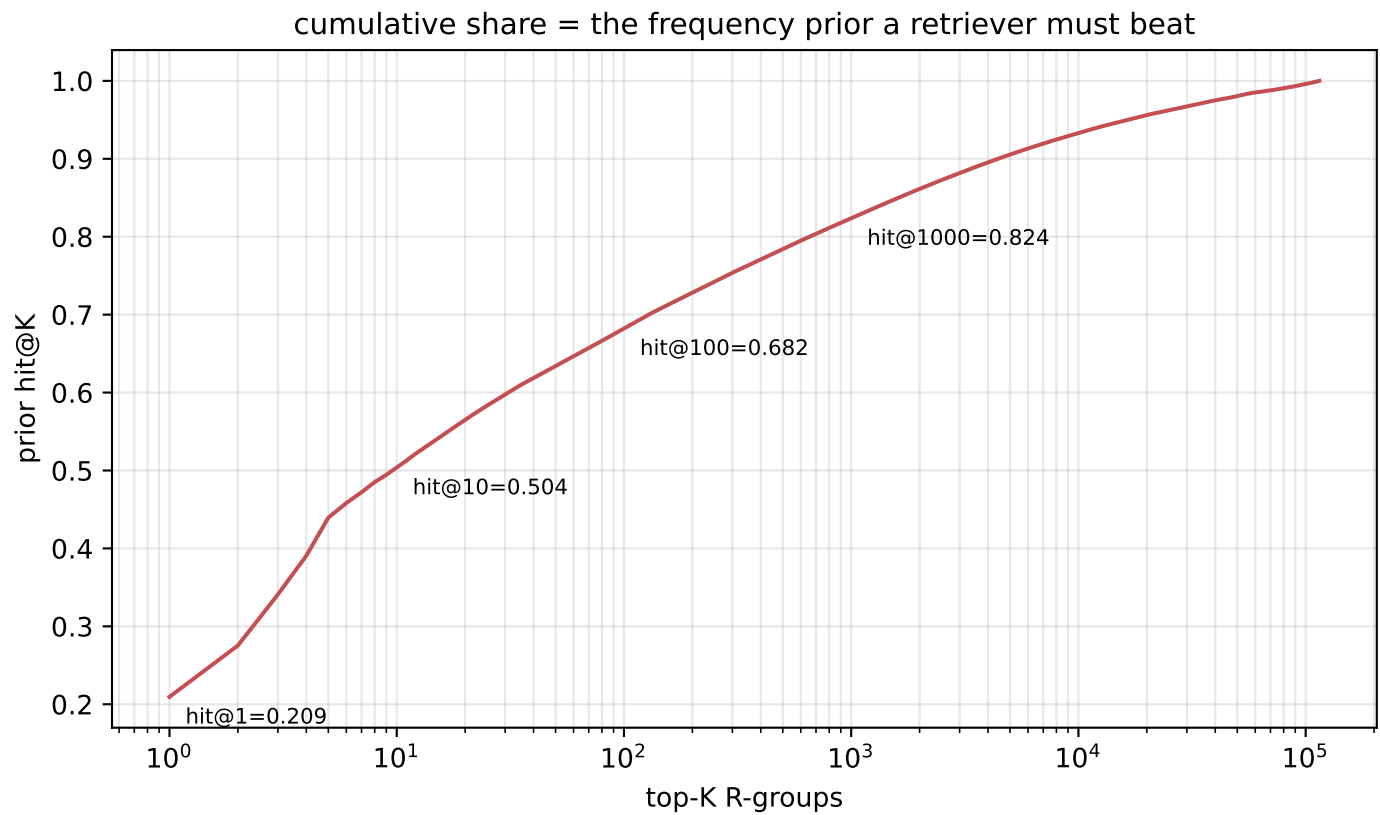
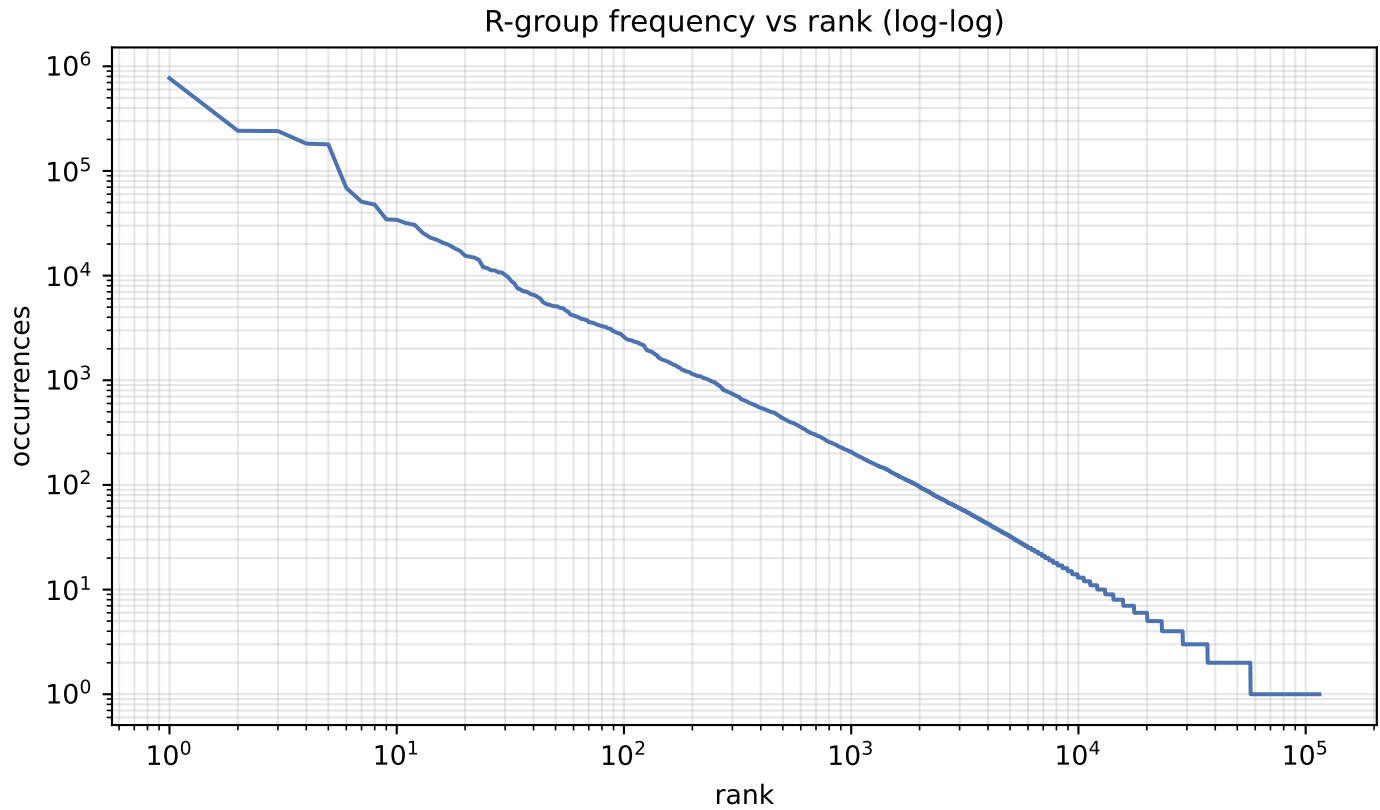


blue = core orange = R-groups red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew



H · R-group library

quantity	value
distinct R-groups	114,816
occurrences	3,670,468
effective size (exp H)	286
top-1 share	20.95%
top-10 cover	50.4%
top-100 cover	68.2%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
768,860	20.95%	*C
241,901	6.59%	*O
240,559	6.55%	*O
182,429	4.97%	*C(C)=O
179,382	4.89%	*OC
68,862	1.88%	*CC
50,781	1.38%	*CO
47,667	1.30%	*c1cccc1
34,420	0.94%	*C(C)C
34,169	0.93%	*OC(C)=O
31,700	0.86%	*CCC
30,542	0.83%	*C(C)C
25,674	0.70%	*C(=O)O
23,146	0.63%	*CCCC
21,992	0.60%	*Cl
20,585	0.56%	*OC
19,606	0.53%	*C(=O)OC
18,206	0.50%	*c1ccc(OC)cc1
17,224	0.47%	*CCCCC
15,471	0.42%	*C(=O)c1cccc1